

Business Briefs By Tom Cullum



A DIFFERENT LOOK AT PRODUCTIVITY

Many a *mau culpa* and much despair has been generated over the question of productivity in the U.S. As in: What is wrong with our work ethic? Or: Why doesn't American management take the long-term view and pump more money into automated procedures? Even though we still have the highest rate of productivity in the world, the rate has been down for several years. And Japan, in particular, has posted a better productivity growth rate than the U.S. for the past 15 years.

It doesn't hurt to be concerned about this. But it doesn't help to get worked up for the wrong reason, either. Which is why Carl Siel of Xerox has made a valuable contribution to the productivity dialogue.

Siel is manager of technical training development, value analysis and value engineering at Xerox's MSE training center. In a paper prepared for delivery this week at Electro/82, he points out that different definitions here and in Japan make it hard to compare productivity measures. Both countries measure productivity as output divided by labor input. In addition to being an outmoded way to measure productivity—since capital, financial and managerial factors should also be considered—this productivity measure tends to obscure a very significant strength in the American economy, Siel points out.

ABSORBING THE BABY BOOM

Comparing the growth of the work force in the U.S. and Japan since 1967 shows that the pool of U.S. workers has grown anywhere between 28 and 37.5 percent. In Japan, the growth was just 4 percent. "While we have been absorbing the (post World War II) baby boom into the work force, Japan has had virtually no increase to absorb," Siel notes. In other words, while output has increased in both countries, the labor input has grown much greater here. So, given the equation of $P = O/L$ (where P = productivity, O = output and L = labor), our relative productivity has, of course, suffered.

As Siel hastens to add, the mathematics of the baby boom doesn't change the fact that the Japanese are narrowing the productivity gap. But, "We should understand that our economy and our environment was strong enough to allow so many jobs as to absorb the baby boom," he explains. "How many other economies (including the Japanese) could have survived a labor onslaught as we have?"

A final note: During the 1980s, the numbers game should benefit the U.S. According to Siel, the American labor force is expected to decline by 16 percent during the current decade, as more people leave the labor force than enter it.

TAPPING NATIONAL'S PRODUCTIVITY PROGRAM

Like many electronics companies, National Semiconductor has embarked on programs to enhance productivity. But the self-described wizards of Silicon Valley have come up with a new trick as well. They plan to sell productivity.

They will do it with TAP, which stands for Targeted Analysis Process. Developed for internal use, TAP is now being offered to other companies as well. Similar to the quality circle tactic, TAP starts with management definition of a problem. A group of three to 10 employees in the affected work area then hash the problem over, often coming up with a solution within the first 60-minute TAP session, according to Tim Thorsteinson, corporate manager for human performance and a co-developer of the program.

"We've got something worth sharing," adds Roy Brant, vice president of human resources. After being used successfully thousands of times at National, TAP has also been tested, with good results, in college, hospital, governmental and service industry environments, he reports.

A TAP kit and training videotape goes for \$250. To tap in, contact National Semiconductor Productivity Products, 2900 Semiconductor Drive, Santa Clara, CA 95051, mailstop 16-293.

STICK WITH THE MAIN EVENT

Northern Telecom chairman A. Jean de Grandpre has some sound advice for any of his Canadian countrymen who might be heartbroken now that an American team (the NY Islanders) has won hockey's prized Stanley Cup three years in a row.

Speaking at last month's annual meeting, de Grandpre noted, "It is of far greater importance to our national future to have a leading, winning, world-class corporation in a key industry and a key technology, than it is to have a winning hockey team." Naturally, he thinks Northern Telecom is that winner. Not only was it the "best-performing high-technology stock listed on the New York and Canadian stock exchanges last year," but "it has established a world leadership in the application of digital technology in the key area of telecommunications switching systems and in the complex technology of custom semiconductors for telecommunications," de Grandpre claims.

Well, maybe he's boasting. But at least he's boasting about the important things.

Newcomers Invade Fault-Tolerant Field, But Tandem Has Big Lead

CUPERTINO, CA — Several new companies are challenging Tandem Computers' overwhelming dominance of the fault-tolerant computer-systems market. But even though some of the upstarts offer products that seem more advanced than the leader's, Tandem officials aren't shaking just yet.

Tandem executives have good reason to shrug off the new competition. For one thing, the total market pie is growing fast enough that Tandem can probably spare a few crumbs, or even slices, for others. And, with a six-year lead on the pretenders, Tandem isn't likely to lose significant business without a major product breakthrough that would negate its learning-curve lead.

Three Who Dare

The newcomers include Stratus Computers (Natick, MA), Synapse Systems (Belmont, MA) and Synapse Computers (Milpitas, CA). Of the three, Stratus seems to be the biggest threat to Tandem's monopoly, having begun shipments in February. Synapse has yet to announce products, much less ship them, and isn't expected to do so for at least another month. August has delivered a few systems this year, but has positioned itself in the industrial-control segment, where transaction-oriented Tandem has not made a big push.

Peter Kastner, manager of marketing development at Stratus, claims that the company has already "taken away some orders from Tandem," selling its system, for example, to a dairy company and a shoe-store chain. While current output of four systems per month is far below what Tandem produces, Kastner argues that Stratus will become more of a heavyweight in the fault-tolerant computer industry because of design features that are superior to the leader's. He also says that Stratus will make inroads based on price, since a minimum-configuration Stratus system costs about \$110,000 less than the \$260,000 price tag he puts on the equivalent Tandem package.

Focusing On Hardware

"Tandem is a software-based system. We're different. We've done it all in hardware," Kastner explains. "Every one of the components we use is self-checking during each operating cycle."

Rather than check in each cycle, Tandem's Guardian software employs periodic review. Parallel processors concurrently operate as main units, and as backups for each other. Intermittently, each CPU informs its fellows that it is available for backup duty, and monitors whether one or more of its fellows is working.

By having processors inform each other when a given set of operations begins and ends, Guardian software also insures that manipulations are not inadvertently repeated—or left hanging in mid data-stream. In addition, Guardian's Encompass program tracks procedures and guarantees that any interrupted activity is resumed from the start.

Synapse will also rely heavily on software, according to Kathryn Dillon, a financial analyst with the Gartner Group in Stamford, CT. But instead of using a parallel-processor approach, Synapse programs will allow one processor to back up several others. This should result in substantial cost savings for many users, she notes.

While the Synapse software approach may be an improvement, incorporating check procedures in hardware is a better way to go, says Stratus spokesman Kastner. That's because users can more easily extend their fault-tolerant operations. "Absolutely no user-software is involved in making existing applications fault-tolerant," he states.

Are More Bits Better?

Kastner also claims that the Stratus approach is superior because it employs 32-bit architecture. While Tandem includes 32-bit addressability in its NonStop II system, internal data paths remain in the 16-bit mode. As Kastner sees it, Stratus has a faster computational ability and a bigger memory system thanks to 32-bit capability.

Tandem marketing director Jerry Peterson doesn't think that 32-bit architecture will make much of a difference. Most NonStop customers are transaction-oriented, and they typically process only small chunks of data in each transaction. Where they need speed is in I/O functions.

The faster speed of 32-bit codes is not as important as in, say, scientific applications, where large number-crunches are involved, Peterson states. And 32-bit addressability gives them all the memory they need—up to a billion k-bytes.

"The bottom line is transaction throughput. Our customers couldn't really care whether you have 16 bits or 32 bits" in internal architecture, he claims.

Whatever the merits of that argument, involved observers like Lorry Roberts, a partner in the investment-banking and venture-capital firm of Hambrecht & Quist, expect Tandem to come out with its own 32-bit product within the next 18 months. "I believe customers will strongly desire 32-bit capability," for fault-tolerant systems, since they will want such systems to be compatible with other computer applications that benefit from the longer codes, he reasons.

As a principal in one of the venture firms that have put up \$15 million to back Stratus, and \$7.9 million to back Synapse, Roberts may be a bit prejudiced about the appeal of 32-bit machines. (Synapse president Mark Leslie had once said that his company wants to produce a 32-bit microcomputer system.) But he doesn't fault Tandem for letting upstart startups take the first plunge into the longer-code systems. "I don't think they've dragged their feet at all on this," he says. "By the time you see volume shipments of 32-bit fault-tolerant machines, don't be surprised if Tandem comes out with one of its own."

Momentum Favors Tandem

Based on projections of market growth, Tandem can probably afford some competition. Looking just at CPU hardware and peripherals specifically packaged with them in fault-tolerant systems, Strategic Business Services, a San Jose-based market-research firm, puts the industry's sales last year at about \$100 million. In a survey published last November, entitled "Survivable Systems: Pitfalls and Opportunities," Strategic projects annual hardware and peripheral sales of \$2.6 billion in just six more years. Counting software sales not included in Strategic's figures, the total could be higher.

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Newcomers C

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gic's analysis, Dataquest forecasts a billion annual volume by 1985.
Given Tandem's sales of \$208 million last year, observers credit it with total domination of the fault-tolerant market. If the Dataquest forecast is correct, Tandem could continue its

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Challenge Tandem In Fault Tolerance

nominal annual growth rate (the company has averaged about 106 percent revenue increases annually for the past three years) but would see its market share drop to about 70 percent. In fact, Tandem management has set its sights on being a \$1 billion company by 1985, which would put its market share at

closer to 20 percent that year). Obviously, someone is going to pick up the slack. Stratus and Synapse clearly hope to grab some of that unclaimed business. They may well get their share, but other would-be pretenders to Tandem's domain will find it tough to operate. "I

don't think there is going to be much more room for a lot of new startups," suggests Strategic's Denis Hoye, director of U.S. client services.

Hoye, who edited his firm's survey of the fault-tolerant industry, predicts that much of the big market ahead will go to traditional computer powers who add higher reliability to their products and then assure total system integrity by distributing more intelligence to terminals via local area networks. With LANs, the loss of a terminal does not affect the operation of the total system.

In particular, 32-bit microcomputers like Intel's iAPX 432, will be "what shortens the opportunities" for Tandem-like companies. The iAPX 432 puts into firmware such attributes as interprocessor communication, processor scheduling, and data protection.

On the other hand, fault-tolerant firmware won't negate the importance of software investment as a determinant of systems selection. Users will continue to look first to vendors who already have sold them considerable software. Unfortunately for the newcomers, this factor would seem to benefit Tandem and other established vendors.

The Establishment Edge

The Securities Industry Automation Corporation (SIAC) in New York City is a case in point. It currently operates several on-line transaction systems where fault-tolerance is a prime concern. In addition to some 70 Tandem systems, SIAC uses IBM and Univac machines to keep computer-crash at a minimum.

SIAC operates the Intermarket Trading System, which links the New York, American and five regional stock exchanges and allows traders to buy and sell 30 different stocks in the most opportune location. This month, the ITS linked up with the NASDAQ over-the-counter market. Later this year, the expanded ITS is supposed to accommodate another 170 stocks.

While living up to these plans will mostly be a software problem, some hardware will probably be added along the way, says John McGee, vp/technical services for SIAC. "We're in the process now of evaluating whether we need more equipment, and we may look at some new vendors," he reports. "But we probably will pretty much go with those vendors in-house."

SIAC's huge investment in software for its current computer arsenal means any newcomer would have to show "a pretty dramatic advantage" over existing vendors to crack the lineup, McGee concludes.

Companies Hope Disks Drive Past Recession

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wards of 45 companies expected in the market by the end of this year, a shake-out will no doubt occur. But the key fact is that many small companies have a chance to succeed because neither the microcomputer makers nor IBM are calling the tune in this market.

"Independents have made considerable progress in setting standards," notes C. Norman Dion, president of Dysan, the dominant firm in high-end microdisks. But while Seagate's Shugart is confident that a standard proposed by his company and Dysan (among others) will win ANSI acceptance, Dion warns against complacency. "Let's not waste the opportunity IBM has provided," he urges.

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